Pierce Bader

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EDUCATION

Columbia University | B.S. Mechanical Engineering | New York, NY **Sept. 2024 – May 2029 (expected)**

- **Relevant Coursework:** Data Science in Python, Intro to Electrical Engineering, Fundamentals of Aerospace Engineering, Multivariable Calculus, Linear Algebra
- **Industry Co-op:** 9 months at Apple in Brittles Manufacturing Design; 3 months at TE Connectivity

Washington Aerospace Scholars | Seattle, WA **Aug. 2022 – June 2023**

Took courses in the University of Washington's aerospace engineering program in collaboration with Boeing and the Seattle Museum of Flight. Selected for summer residency. Developed a plan for a manned mission to Mars.

EXPERIENCE

Apple **| Cupertino, CA | Manufacturing Quality Engineering Intern** **Jan. 2026 – Sept. 2026**

- Took ownership as the lead responsible quality engineer for 2 NPI components in the portables LOB
- Enabled over 20% yield improvement to beat legacy FPYs on new generation brittles components; incorporating traceability and providing supply chain transparency
- De-risked 3 critical mass production blockers, enabling an aggressive NPI shipping schedule and ensuring seamless product launches while bringing up new metrology and inspection capabilities
- Worked closely with commercial and process teams to enable new product features, metrology equipment, and cost savings on MacBook programs
- Drove critical supplier, mechanical, and reliability quality issues to closure across 4 LOBs and 6 NPI programs
- Led quality analysis and build reviews, highlighting process excursions and driving cross-functional alignment on yield improvement actions across internal teams and external vendors

TE Connectivity **| Harrisburg, PA | Aerospace Engineering Intern** **May 2025 – Aug. 2025**

- Saved $370,000 annually by reducing new product time-to-market in the NPI Lab
- Implemented Six Sigma statistical analysis techniques to perform measurement system analyses, design of experiments studies, and capability studies using Minitab, Python, and Keyence optical inspection systems
- Optimized 3D printing cell processing workflows for Stratasys Origin, Fortus, and BMF MicroArch printers; used Creo to design DoE benchmarks for dimensional accuracy across platforms

Columbia Airplane Club **| New York, NY | Chief Quality Control Engineer** **Sept. 2024 – Jan. 2026**

- Constructed a UAV that ranked 16th out of 112 teams at the 2025 Design, Build, Fly in Tucson, Arizona
- Led the electrical sub-team in the hardware and software development of a GPS homing glider
- Used SolidWorks and Fusion to develop robust payload attachment and hardpoint solutions
- Elected as a member of the executive board to lead 30 engineers through a national UAV design competition

ACTIVITIES & PROJECTS

Talon 1400 Unmanned FPV Drone **| New York, NY | Project Lead** **Sept. 2025 – Present**

- Utilizing additive manufacturing to build a Flightory Talon 1400
- Designed an ArduPilot-based autopilot

SSBUR Home **| New York, NY | Manufacturing Intern** **Sept. 2025 – Jan. 2026**

- Worked with a Brooklyn-based product designer and engineer to manufacture their home goods products

SKILLS & INTERESTS

- **Engineering:** Fusion, SolidWorks, Creo, OnShape, GD&T, DLP & FDM 3DP, Laser Cutting, DoE, MSA, JMP, ORT/Rel Analysis, Process Capability Analysis, Minitab
- **Computer Science:** Python, Java, C++, Swift, JASP, HTML
- **Interests:** Club Tennis, Club Sailing, OSINT, FPV Drones